

9. Why is the sum of a number with a 3 in the ones place and a number with a 4 in the ones place always an odd sum?
10. How can you explain why the digits in the ones place of the addends determine if the sum will be even or odd?

11. **STEM Connection** A baking sheet can hold an even or odd number of pastries. If Saffron wants an even number of pastries, how can she arrange the pastries on 2 sheets?



12. **Extend Your Thinking** Carol designs scarves. If she uses 112 blue, 113 yellow, and 114 brown stripes on a scarf, will the scarf have an even or odd number of stripes? Explain how you know.

Reflect

How can addition patterns help you justify that a sum is correct?

Math is... Mindset

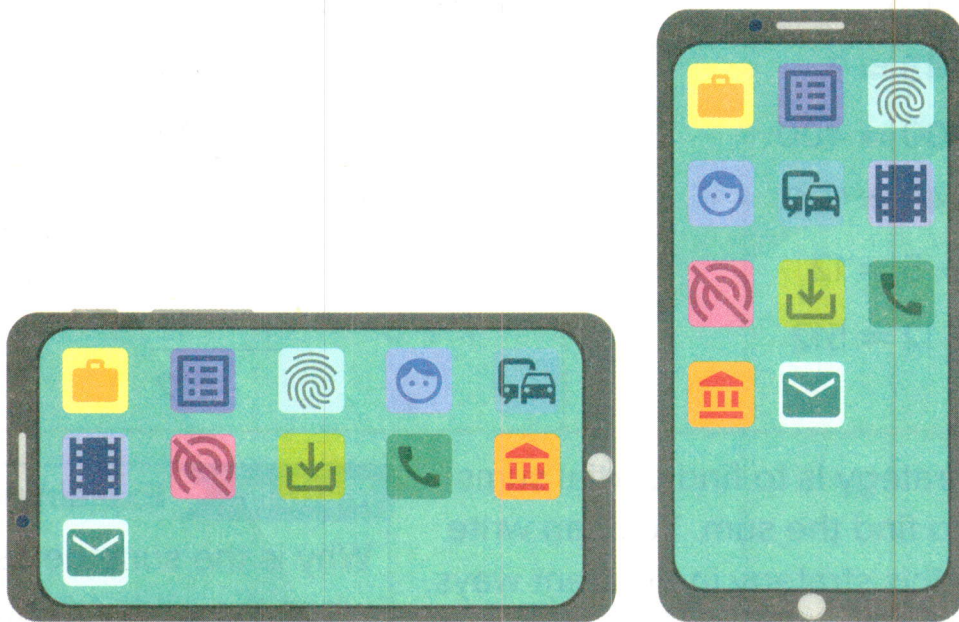
How did you use strategies to work more efficiently?

Use Partial Sums to Add



Be Curious

How are they the same?
How are they different?



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Math is... Mindset

What are your strengths in math?

Learn

Maggie and Tanesha break apart the addends by place value to add $367 + 145$. They show their work in different ways.

How can each girl show their work in a different way?

hundreds	tens	ones
3	6	7
1	4	5

You can **decompose**, or break apart, the addends by place value to find partial sums. Then, add the partial sums to find the sum.

► **One Way** You can write the addends in a row.

$$\begin{aligned}
 367 + 145 &= ? \\
 300 + 100 &= 400 \\
 60 + 40 &= 100 \\
 7 + 5 &= 12 \\
 400 + 100 + 12 &= 512
 \end{aligned}$$

► **Another Way** You can stack the addends.

$$\begin{array}{r}
 367 \\
 + 145 \\
 \hline
 512
 \end{array}$$

partial sums

One addition strategy is to find partial sums and add them to find the sum. You can write the addends in this strategy in different ways.

Math is... Explaining

Why is the sum the same when the addends are in a row or stacked?

Work Together

Ari's work is shown. Jun sees Ari's work and says 500 is the sum of $309 + 225$. Do you agree? Explain your reasoning.

$$\begin{array}{r}
 309 \\
 + 225 \\
 \hline
 534
 \end{array}$$

On My Own

Name _____

How can you decompose each addend? What is the sum?

1. $247 + 564 =$ _____

2.
$$\begin{array}{r} 815 \\ + 148 \\ \hline \end{array}$$

3.
$$\begin{array}{r} 729 \\ + 148 \\ \hline \end{array}$$

4. $327 + 176 =$ _____

5. Whitney uses partial sums _____ + _____ = 598 to add. Look at her work to determine which two numbers were in her original equation.

$$200 + 300 = 500$$

$$10 + 80 = 90$$

$$6 + 2 = 8$$

$$500 + 90 + 8 = 598$$

6. How can you determine which addends are in the original equation by looking at the partial sums?

7. Tyrone spent 172 days in school last year. If he attends school the same number of days next year, how many days will he spend in school in two years?

How can you find the sums in a different way?

8. $475 + 325 = 800$
 $400 + 300 = 700$
 $70 + 20 = 90$
 $5 + 5 = 10$
 $700 + 90 + 10 = 800$

9.
$$\begin{array}{r} 238 \\ +271 \\ \hline 400 \\ 100 \\ + 9 \\ \hline 509 \end{array}$$

10. Eleanor's watch shows her steps before lunch. Then she took 486 more steps. How many total steps did she take?



11. **Error Analysis** Amal adds $378 + 141$. She decomposes each number and adds $300 + 100 = 400$. She writes $378 + 141 = 400$. How can you explain her mistake?

12. **Extend Your Thinking** How can you solve $249 + 401 + 276$ using partial sums? Show your work.

Reflect

Why is understanding place value important when using the partial sums strategy to add?

Math is... Mindset

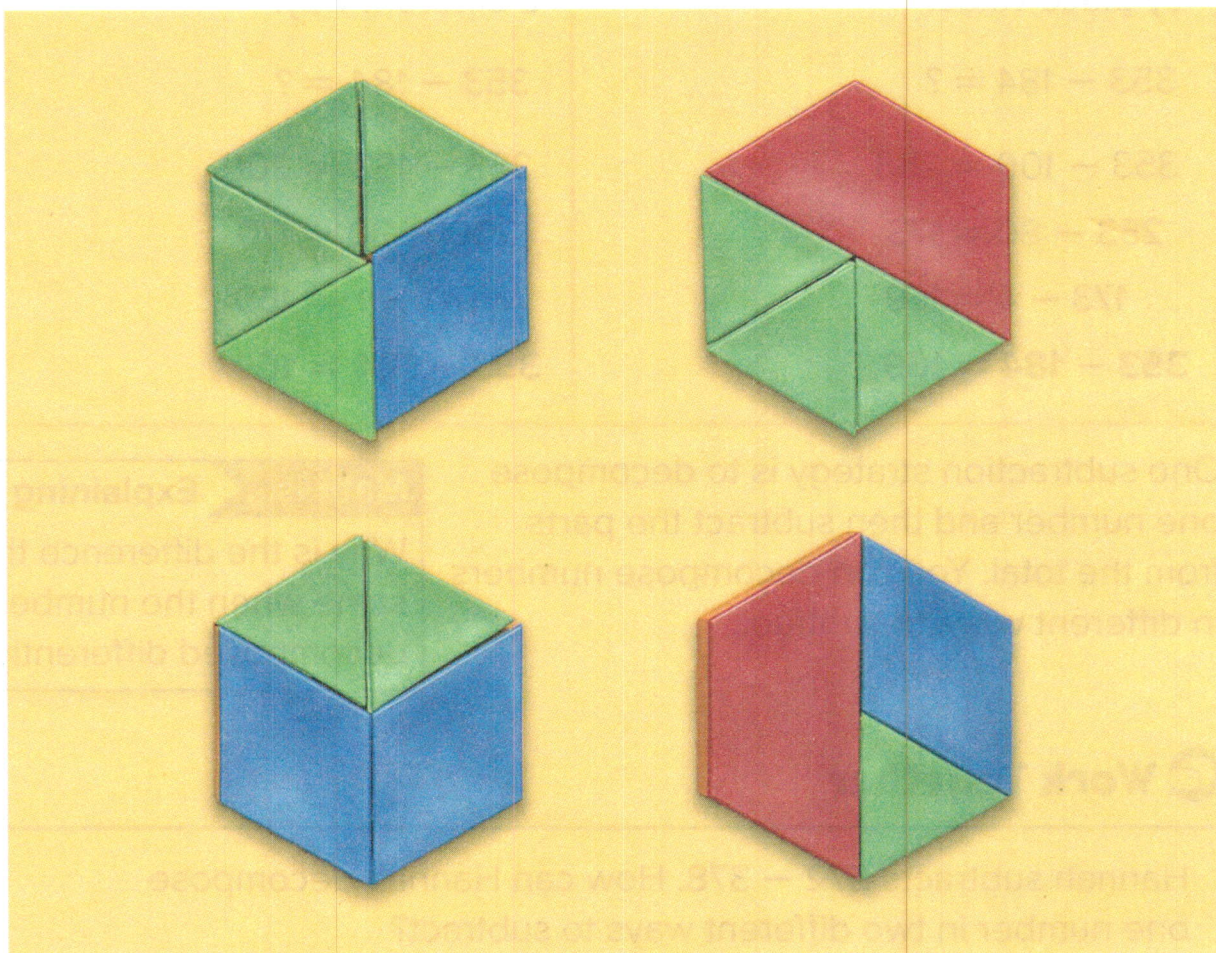
What strengths did you use today? What can you work on?

Decompose to Subtract



Be Curious

How are they the same?
How are they different?



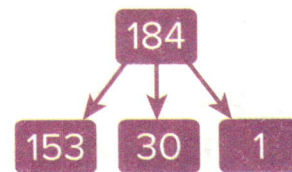
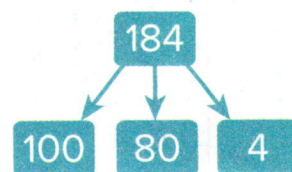
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Math is... Mindset

How can creative thinking help you solve a problem?

Learn

Samir and Jen subtract $353 - 184$. Samir decomposes one number to subtract. Jen decomposes the same number in a different way to subtract.



Will Jen find the same difference as Samir?

► **One Way** You can decompose one number by place value.

$$353 - 184 = ?$$

$$353 - 100 = 253$$

$$253 - 80 = 173$$

$$173 - 4 = 169$$

$$353 - 184 = 169$$

► **Another Way** You can decompose one number in a different way.

$$353 - 184 = ?$$

$$353 - 153 = 200$$

$$200 - 30 = 170$$

$$170 - 1 = 169$$

$$353 - 184 = 169$$

One subtraction strategy is to decompose one number and then subtract the parts from the total. You can decompose numbers in different ways to subtract.

Math is... Explaining

Why is the difference the same when the number is decomposed differently?

Work Together

Hannah subtracts $572 - 378$. How can Hannah decompose one number in two different ways to subtract?

On My Own



Name _____

How can you decompose the number in 2 ways?

1. 367

2. 876

How can you decompose one number to subtract? Why did you choose that way?

3. $495 - 122$

4. $639 - 370$

How can you find the difference? Show the strategy you used.

5. $284 - 182 =$ _____

6. _____ $= 333 - 114$

7. $502 - 382 =$ _____

8. $744 - 466 =$ _____

9. Error Analysis Juan subtracts $345 - 101$. He decomposes 101 into 100 and 10 and subtracts the parts from 345. How can you help him understand his mistake?

10. The table shows the number of people who attended the school fun fair each day. Show a strategy to find the difference between the greatest and least number of people.

Fun Fair Visitors	
Day	Number of People
Thursday	103
Friday	168
Saturday	257
Sunday	224

11. A baker bakes 268 bread rolls. 155 are cinnamon rolls. The rest are plain rolls. How many plain rolls does she bake?

12. Extend Your Thinking Ana subtracts $438 - 342$ by decomposing 342. She subtracts 2, then 300, and finally 40. Can she subtract the parts in any order? Explain your reasoning.

Reflect

How do you decide how to decompose one number to subtract?

Math is... Mindset

How did creative thinking help you solve a problem?

Adjust Numbers to Add or Subtract



Be Curious

Which doesn't belong?

$$38 + 19$$

$$40 + 17$$

$$37 + 20$$

$$32 + 25$$

Math is... Mindset

How can working as a team help you achieve your goal?

Learn

There are 224 Tigers fans and 212 Hawks fans at the town football game. 109 more Tigers fans arrive late to the game.

How many more Tigers fans than Hawks fans are at the game?



Adjust Addition Equations

$$224 + 109 = ?$$

-1

+1

$$223 + 110 = 333$$

Subtract from one addend and add that amount to the other.

Adjust Subtraction Equations

$$333 - 212 = ?$$

-2

-2

$$331 - 210 = 121$$

Subtract or add the same amount to both numbers.

You can adjust numbers to get numbers that are easier to work with. Adjust both numbers to keep the sum or difference the same as the original.

Math is... Structure

How is adjusting a subtraction equation different from adjusting an addition equation?

Work Together

Marco adds $457 + 208$. He adjusts 457 to 460 and 208 to 211. He finds the sum 671. Do you agree with his strategy? Explain.

On My Own



Name _____

How can you adjust the equation to solve?

1. $272 - 99 =$ _____

2. $139 + 216 =$ _____

3. $847 - 332 =$ _____

4. $398 + 147 =$ _____

5. $491 - 226 =$ _____

6. $129 + 139 =$ _____

7. Alex adjusted the equation $109 + 119 = ?$ to $110 + 118 = ?$.
Describe how he adjusted the equation. Why do you think he did it in that way?

8. Show how you can adjust $396 - 226$. Use a number line to show that the new equation has the same difference.

9. You can adjust $236 - 119$ different ways. How can you explain one way you can adjust and why the equation is easier?

10. Melissa and Juan are finding $129 + 257$ by adjusting. Melissa tries solving using $130 + 258 = 388$ and Juan solves it using $130 + 256 = 386$. Which sum is correct? Explain.

11. **STEM Connection** Saffron completed 851 orders of pastries her first year. After her second year, she completed 926 orders. How many more pastry orders did she complete in her second year than in her first year?



12. **Extend Your Thinking** Yazmin adds $457 + 251$. She adjusts 251 to 250, but forgets to adjust 457. She adds $457 + 250 = 707$. How can she adjust the sum to fix her mistake? Explain your reasoning.

Reflect

How can adjusting just one number affect the addition or subtraction equation?

Math is... Mindset

How have you worked as a team to achieve your goal?

Use Addition to Subtract



Be Curious

What do you notice?
What do you wonder?



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Math is... Mindset

How can you show you understand
how others are feeling?

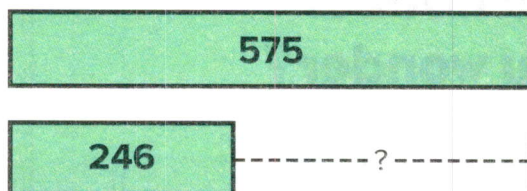
Learn

Adaline and her family are driving 575 miles to visit her grandmother. They drive 246 miles and stop for lunch.



How can you find how many miles they have left to drive?

You can represent the problem with a **bar diagram**.



You can write a subtraction equation with an unknown difference.

$$575 - 246 = ?$$

$$575 - 246 = \mathbf{329}$$

You can write an addition equation with an unknown addend.

$$246 + ? = 575$$

$$246 + \mathbf{329} = 575$$

Addition and subtraction are related. You can rewrite a subtraction equation as an addition equation with an unknown addend.

Math is... Relationships

How can using an unknown addend equation be helpful when you subtract?

Work Together

Deigo wants to use addition to solve the equation $478 - 326 = ?$. How can you explain this strategy?

On My Own

Name _____

How can you write a related addition equation for the subtraction equation?

1. $635 - 202 = ?$ _____

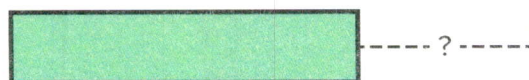
2. $400 - 151 = ?$ _____

3. $825 - 134 = ?$ _____

4. **Error Analysis** Jayla knows she can use addition to subtract. She adds $301 + 447$ to find the difference of $447 - 301$. How can you help her understand how to use this strategy?

5. On Monday, 549 people visited the aquarium. On Friday, 823 people visited the aquarium. How many more people visited the aquarium on Friday?

- a. How can you fill in the bar diagram to represent the problem?



- b. How can you write an equation with an unknown to represent the bar diagram?

How can you use the relationship between addition and subtraction to find the difference?

6. $480 - 318 =$ _____

7. _____ $= 300 - 179$

8. $705 - 239 =$ _____

9. $212 - 135 =$ _____

10. Evan and Kyle put together a 500-piece puzzle. Evan placed 247 pieces. How many pieces did Kyle place?

11. **STEM Connection** Saffron's Bake Shop can spend \$700 to buy a large and a small mixer. She spends \$411 on the large mixer. How much does she have left to spend on a small mixer?



12. **Extend Your Thinking** Kwan says that he can solve $445 - ? = 239$ by using the equation $445 + 239$. Do you agree? Explain why.

Reflect

How are addition and subtraction related?

Math is... Mindset

How did you show you understand how others are feeling?

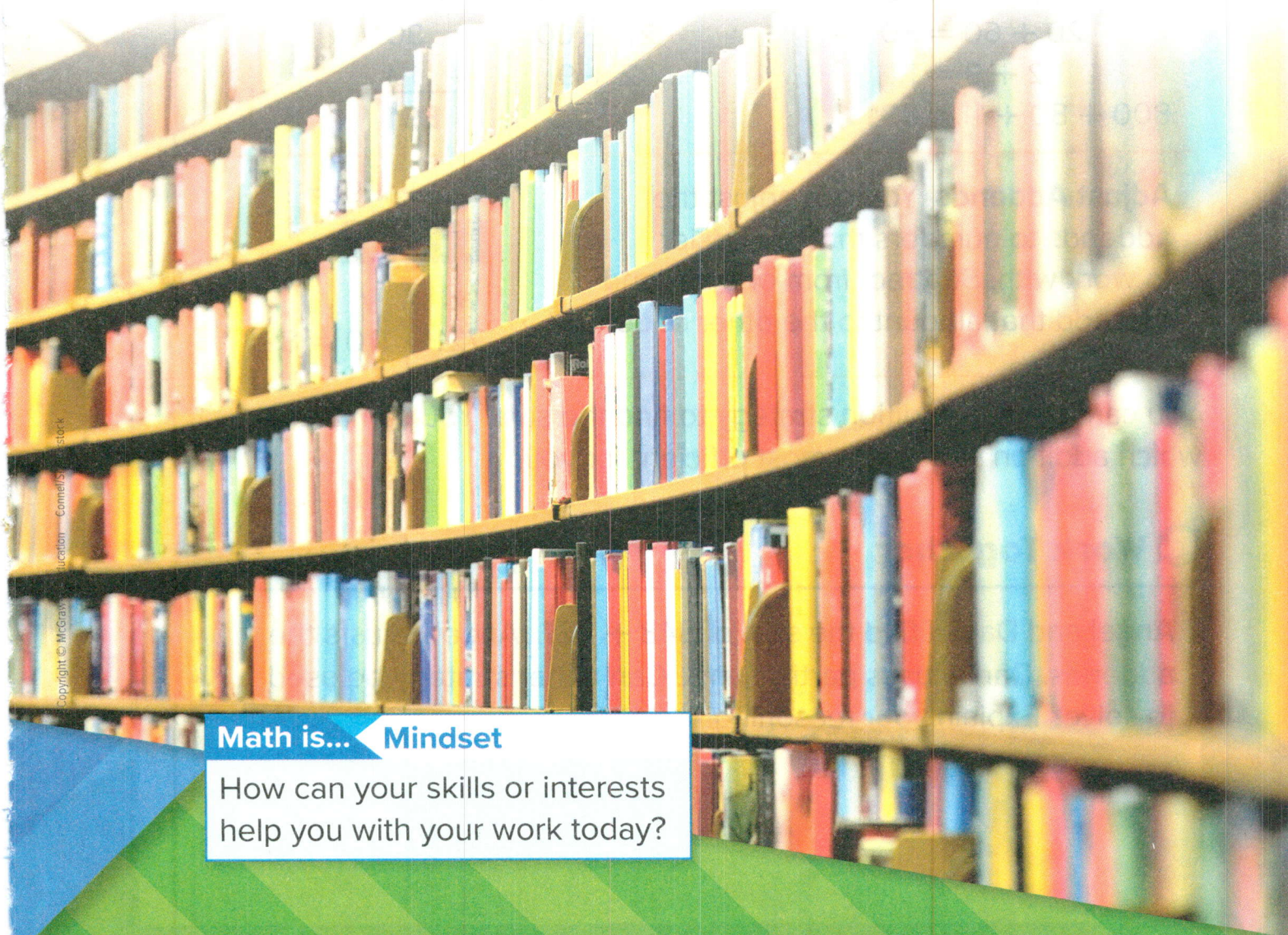
Fluently Add within 1,000



Be Curious

What question could you ask?

Last year the library's collection of books increased.
This year additional books were added to the library.



Math is... Mindset

How can your skills or interests help you with your work today?

Learn

Last year the library's collection of books grew from 350 to 722 books. This year 169 books were added to the library.

What strategies can you use to find the total number of books in the library?

Math is... Exploring

How can you decide which information in the problem is useful?

Partial Sums You can decompose each addend by place value and add the place values to find partial sums. Then add the partial sums to find the total.

$722 + 169 = ?$	
$700 + 100 = 800$	$700 + 100$
$20 + 60 = 80$	$20 + 60$
$2 + 9 = 11$	$2 + 9$
$800 + 80 + 11 = 891$	

$$\begin{array}{r} 722 \\ +169 \\ \hline 800 \\ 80 \\ + 11 \\ \hline 891 \end{array}$$

Adjust Addends You can adjust the addends to numbers that are easier to work with. Subtract from one addend and add that amount to the other addend.

$$\begin{array}{r} 722 + 169 = ? \\ \begin{array}{cc} \boxed{-1} & \boxed{+1} \\ \downarrow & \downarrow \end{array} \\ 721 + 170 = 891 \end{array}$$

You can use any addition strategy to find the sum. You can decide which addition strategy is most efficient for you.

Work Together

Angel and Daniel find the sum of $348 + 227$. Angel adjusts the addends. Daniel uses partial sums. Can either strategy be used to solve? Explain your reasoning.

On My Own

Name _____

How can you solve the equation?

1. $437 + 269 =$ _____

2. _____ $= 123 + 443$

3. _____ $= 367 + 512$

4. $791 + 111 =$ _____

5. **Error Analysis** Priya is adding $327 + 478$. Priya adds 3 to 327 to add $330 + 478$. Then she subtracts 3 from the sum. Do you agree with her strategy? Explain.

6. Sam adds $249 + 281$ by writing his addends in a row. What is another way Sam can show $249 + 281$?

7. Max's little brother Henry is 2 years old. There are 365 days in a year. How many days old is Henry?

How can you solve the equation? Explain your addition strategy.

8. $458 + 139 =$ _____

9. $719 + 234 =$ _____

10. Mr. Perez needs 650 sheets of construction paper for his class art project. How many packs of this construction paper should he buy?



11. Tryon School raised \$345 at their school fundraiser and \$486 at their bake sale. They used \$650 of the money they raised to buy new playground equipment. How much money did Tryon School raise?

12. **Extend Your Thinking** The community center had 113 swimmers in the fall class and 235 swimmers in the winter class. The summer class had 105 more swimmers than both classes combined. How many swimmers were in the summer class?

Reflect

How do you decide which addition strategy to use when solving a problem?

Math is... Mindset

How did you use your skills or interests to help you with your work?

Fluently Subtract within 1,000



Be Curious

Is it always true?

Niko says the only way to subtract $348 - 135$ is to decompose 135.



$$348 - 135 = ?$$

$$348 - 100 = 248$$

$$248 - 30 = 218$$



$$218 - 5 = 213$$

Math is... Mindset

What goal do you want to accomplish today?

Learn

Mateo sold 543 tickets to the newest action movie Friday night. He sold 134 fewer tickets on Saturday night than Friday. Then Sunday afternoon he sold 248 tickets.

What strategies can you use to find the number of tickets he sold Saturday?

Math is... Exploring

How is this problem like other problems you have solved?

Decompose One Number

You can decompose one number in different ways and subtract the parts.

$$543 - 100 = 443$$

$$443 - 30 = 413$$

$$413 - 3 = 410$$

$$410 - 1 = 409$$

$$543 - 134 = \mathbf{409}$$

Adjust Numbers

You can adjust the numbers to numbers that are easier to work with. Subtract from or add the same amount to both numbers.

Handwritten work for Adjust Numbers:

$$\begin{array}{r} 550 \\ 543 \\ +7 \\ \hline 550 \end{array}$$
$$543 - 134 = ?$$
$$\begin{array}{cc} +6 & +6 \\ \downarrow & \downarrow \\ 549 & -140 = 409 \end{array}$$
$$134 + 7 = 141$$
$$550 - 141 = 409$$

Related Addition Equation

You can write an addition equation with an unknown addend to find the difference.

$$543 - 134 = ?$$

$$543 - 134 = \mathbf{409}$$

$$134 + ? = 543$$

$$134 + \mathbf{409} = 543$$

You can use any subtraction strategy to find the difference. You can decide which subtraction strategy is most efficient for you.

Work Together

What strategy might be useful to solve $542 - 118$?
Explain your reasoning.